Final project manuscript

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Teach the evidence-to-conclusion chain for Final project manuscript

Preserve failures, grouping and missing outcomes consistently across abstract, tables, figures and artifacts.

Keep one habit: Model the main case → reveal the picture gradually → probe limits → assess independent reasoning → plan a changed-case return.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of final project manuscript

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Independent reproduction check. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of independent reproduction check

Reference example, not the only acceptable wording: Teacher explanation: Independent reproduction checks whether another operator can use the recorded artifacts to recover the stated computation without hidden help.

Provide manifest, fixture data, code, environment and expected metric definition. Record assistance and mismatches; reproduction is not new-data replication.

Without opening the old lesson, explain one key idea from Writing results and limitations. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of writing results and limitations

Reference example, not the only acceptable wording: Teacher explanation: Results and limitations should state observed estimates before interpreting them. Separate measured comparisons from causal or deployment claims.

Give counts, denominators, effect sizes, valid uncertainty and failed conditions. State missing **evidence** and preserve the original primary outcome.

Without opening the old lesson, explain one key idea from Negative results and claim revision. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of negative results and claim revision

Reference example, not the only acceptable wording: Teacher explanation: Negative results report that the **evidence** did not meet a declared expectation.

They can reveal design limits or useful constraints without proving no effect anywhere. Compare results with the original criterion and uncertainty assumptions.

Preserve the failed hypothesis, discuss plausible **mechanism**s and label revised questions as new work.

Pictured end goal

Explain final project manuscript using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Independent reproduction check

Writing results and limitations

Negative results and claim revision

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. A final manuscript integrates question, methods, results, limitations and reproducibility. Each table and claim should point to the correct run.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of final project manuscript

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model the whole case and check each inference

> Keep the independent answer private until after the first attempt.

Definitions and complete contract

A final manuscript integrates question, methods, results, limitations and reproducibility. Each table and claim should point to the correct run.

Audit consistency across abstract, figures, raw results and artifact manifest. Remove unsupported universal claims; label exploratory analyses and unassessed outcomes.

Manuscript item	Complete supplied exercise contract
Main raw result	Eighteen passes among twenty evaluated cases; two failures
Main inconsistency	Abstract says 100%; figure omits two failures
Required main correction	90% over twenty with two failures visible throughout
Separate picture	A correct 7/10, B 8/10; all photos indoor
Picture grouping	Five plants, two views each; ten photos total
Independence evidence	Five identities are not proof of independent sampling
Artifact linkage	Each claim points to its actual run, data and version; do not invent identifiers

Manuscript item	Complete supplied exercise contract
Unknown properties	Population superiority, significance, outdoor performance and detailed case outcomes

Main evidence ledger and checked key

Main manuscript location	Current inconsistency	Supported correction
Raw results table	18/20	Keep original twenty-case record
Abstract	100% success	18/20 = 90% on these evaluated cases
Figure	Two failures absent	Restore both failures and total twenty
Failure count	Hidden by headline/figure	Two failures explicitly reported
Run/artifact references	Need consistency	Identify the actual linked record; no fabricated IDs
Broader claim	Not established	Keep case/outcome scope attached

Main eighteen of twenty is 90%, with two failures; align abstract, figure, table and actual run record. Picture A 7/10 = 70%, B 8/10 = 80%, gain one photo or ten percentage points. All photos are indoor, from five plants with two views each. Correct ten-independent-plants wording to the actual source grouping, while still requiring sampling independence justification; no invented plant-level rate.

Independent twelve eligible records include ten completed, two missing and eight passes: completed-case 8/10 = 80%, coverage 10/12 ≈ 83.333%, two unassessed and two completed failures. The 8/12 = 66.667% known-pass/all-eligible rate is different. Do not call all twelve evaluated or claim population superiority.

Final project manuscript — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Abstract says 100% success; results table shows 18/20, and two failures are omitted from the figure.	Audit consistency across abstract, figures, raw results and artifact manifest. Remove unsupported universal claims; label exploratory analyses and unassessed outcomes.	Correct to 90% over 20 with two failures; restore figure and align abstract/table/run. Do not hide failures to preserve the headline.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Correct to 90% over 20 with two failures; restore figure and align abstract/table/run. Do not hide failures to preserve the headline. Qualification: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify **evidence**. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Abstract says 100% success; results table shows 18/20, and two failures are omitted from the figure.

Correct to 90% over 20 with two failures; restore figure and align abstract/table/run. Do not hide failures to preserve the headline.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Boundary and misconception probes

Manuscript audit	Checked interpretation
Abstract 100%, raw 18/20	Correct headline to 90%; retain two failures
Figure hides failures	Restore them and denominator, not just a caveat
One manuscript has multiple runs	Point every table/claim to the correct version
Ten photos from five plants	Report repeated source views
Plant IDs are distinct	Not automatic sampling independence
Raw case-level outputs absent	Do not invent transitions or a plant-level score
Two missing cases	Report coverage and unassessed status, not evaluated-as-complete
Polished manuscript	Does not create evidence or erase limitations

Separate pictured facts and full reasoning

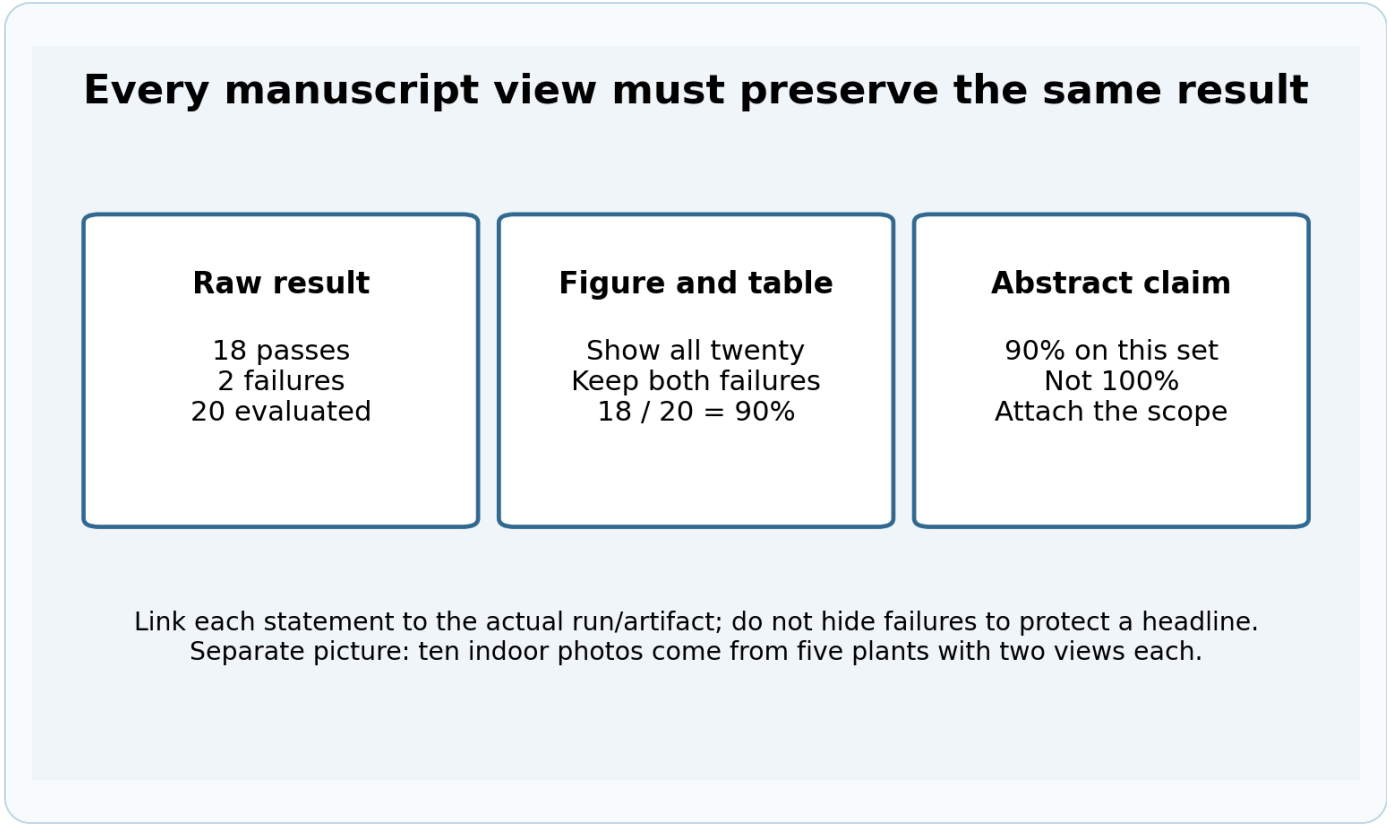
Build a manuscript from claim to record. A capstone manuscript compares two frozen pattern classifiers on ten reserved cases.

A is correct on 7/10; B on 8/10. All cases are indoor, from five plants with two views each.

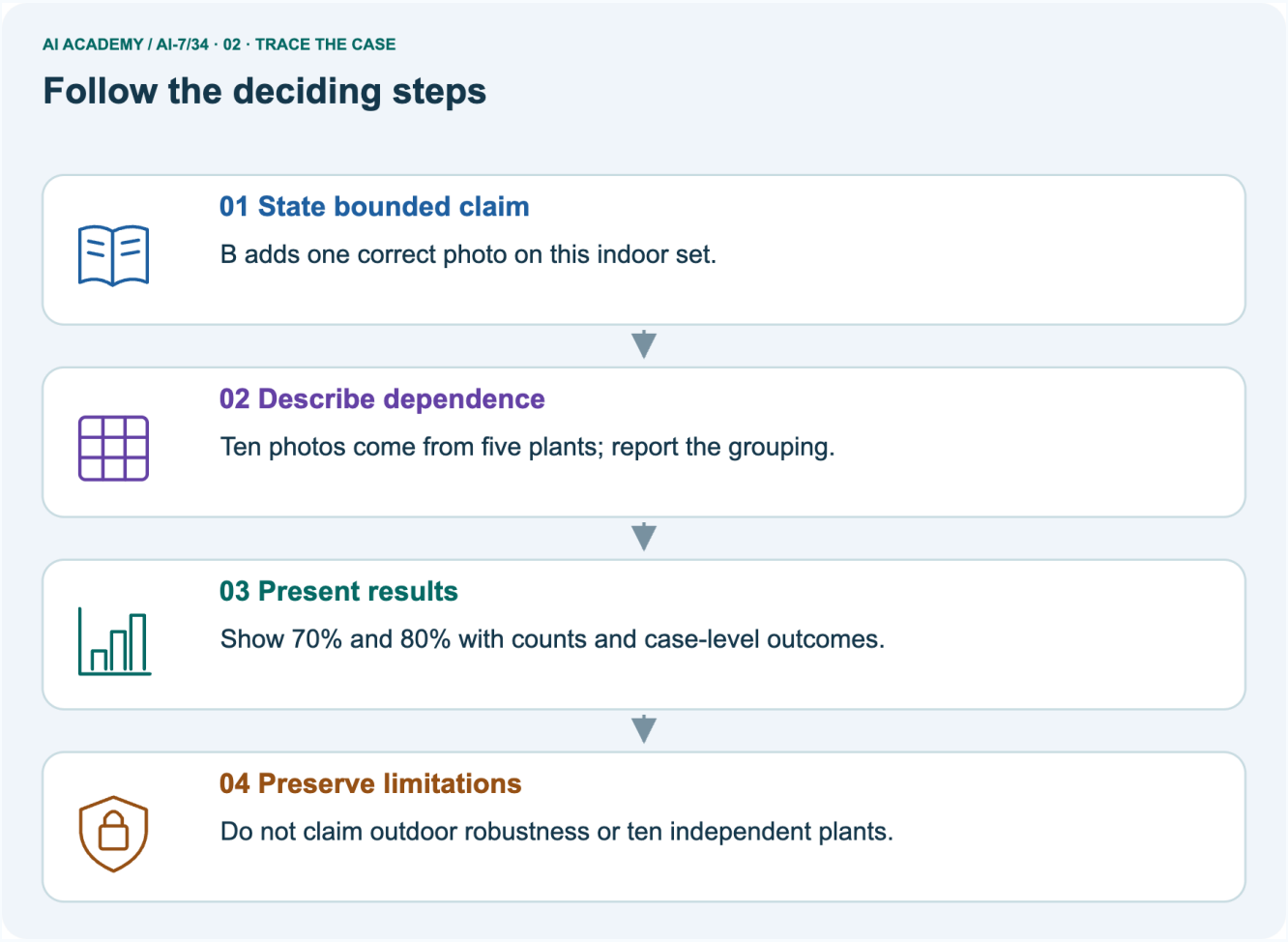
The separate capstone picture compares A seven correct of ten photos and B eight correct of ten: 70% and 80%, a one-photo or ten-percentage-point observed difference. Those ten indoor photos come from five plants with two views each. They are not ten independent plants, and five source identities alone do not establish independent sampling. Report grouping and use analysis assumptions justified for the actual design.

No per-case output table is supplied here, so do not invent which photos changed or a plant-level score. A final manuscript can report the supplied photo-level counts while clearly acknowledging indoor coverage, repeated views and absent population/significance evidence.

Separate capstone record	Supported value or limit
A photo correctness	7/10 = 70%
B photo correctness	8/10 = 80%
Observed difference	One correct photo; ten percentage points
Source grouping	Five plants × two views = ten photos
Setting	Indoor only
Ten independent plants claim	Wrong count and unsupported independence premise
Five plant identities	Correct source count; sampling independence still needs justification
Case-level and plant-level outcomes	Not supplied; do not reconstruct them
Outdoor / population / significance	Not established



Picture trace and changed-condition key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. State bounded claim	B adds one correct photo on this indoor set.
2. Describe dependence	Ten photos come from five plants; report the grouping.
3. Present results	Show 70% and 80% with counts and case-level outcomes.
4. Preserve limitations	Do not claim outdoor robustness or ten independent plants.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A final manuscript aligns question, procedure, **evidence** and scope. Every figure and conclusion should point back to the records that support it.

No population superiority or significance claim is supplied; dependence and sample coverage need explicit treatment.

Change: The abstract calls the study a test on ten independent plants.

Prediction: Correct it to five plants with two views each.

Reason: Five source plants have two views each; independence requires a justified sampling design.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Manuscript audit	A consistency check across research records and prose
Run linkage	Connecting reported numbers to exact results
Evidence alignment	Agreement between claims and their supporting artifacts

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

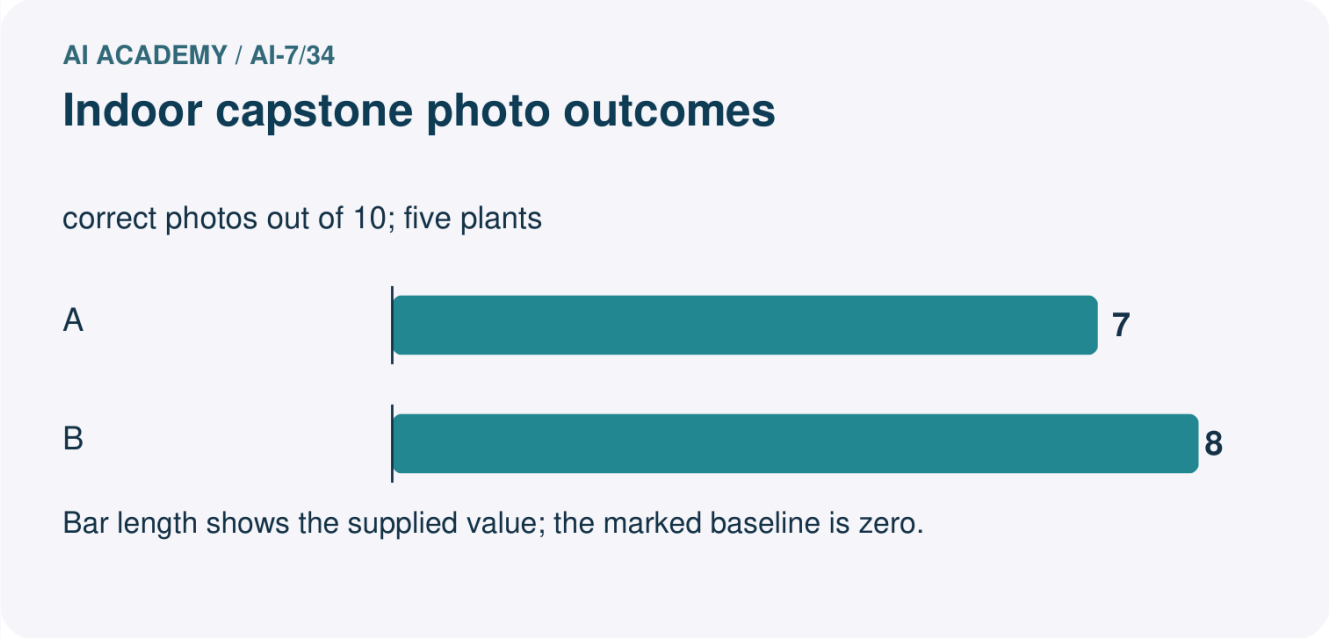
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Final project manuscript

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/34

Final project manuscript

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State bounded claim

B adds one correct photo on this indoor set.



2. Describe dependence

Ten photos come from five plants; report the grouping.



3. Present results

Show 70% and 80% with counts and case-level outcomes.



4. Preserve limitations

Do not claim outdoor robustness or ten independent plants.

CHANGE ONE THING

The abstract calls the study a test on ten independent plants.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Abstract says 100% success; results table shows 18/20, and two failures are omitted from the figure. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Correct to 90% over 20 with two failures; restore figure and align abstract/table/run. Do not hide failures to preserve the headline.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Final project manuscript. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify **evidence**.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Connect the manuscript's 7/10 and 8/10 scores to the five-plant grouping. Which abstract claim must be corrected if it says ten independent plants?

B adds one correct photo on this indoor set. Ten photos come from five plants; report the grouping. Show **70%** and **80%** with counts and case-level outcomes. Do not claim outdoor robustness or ten independent plants. Changed case: Correct it to five plants with two views each. Five source plants have two views each; independence requires a justified sampling design.

Changed-case reasoning: Correct it to five plants with two views each. Five source plants have two views each; independence requires a justified sampling design.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main eighteen of twenty is 90%, with two failures; align abstract, figure, table and actual run record. Picture A 7/10 = 70%, B 8/10 = 80%, gain one photo or ten percentage points. All photos are indoor, from five plants with two views each. Correct ten-independent-plants wording to the actual source grouping, while still requiring sampling independence justification; no invented plant-level rate.

Independent twelve eligible records include ten completed, two missing and eight passes: completed-case 8/10 = 80%, coverage 10/12 ≈ 83.333%, two unassessed and two completed failures. The 8/12 = 66.667% known-pass/all-eligible rate is different. Do not call all twelve evaluated or claim population superiority.

Diagnose the current misconceptions

The worked result establishes every future case: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify **evidence**.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Manuscript says 12 evaluated cases, raw record has 10 completed and 2 missing with 8 passes. Correct the summary.

Teacher explanation: 8/10=**80%** completed pass, coverage10/12 and 2 unassessed; state all denominators. Do not imply12 observed outcomes.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A final manuscript integrates question, methods, results, limitations and reproducibility. Each table and claim should point to the correct run.

Audit consistency across abstract, figures, raw results and artifact manifest. Remove unsupported universal claims; label exploratory analyses and unassessed outcomes.

W2 Trace the worked case

Abstract says 100% success; results table shows 18/20, and two failures are omitted from the figure. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Correct to 90% over 20 with two failures; restore figure and align abstract/table/run. Do not hide failures to preserve the headline.

W3 Audit the limit

A report omits this qualification: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify **evidence**.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify **evidence**.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify **evidence**.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Correct to 90% over 20 with two failures; restore figure and align abstract/table/run. Do not hide failures to preserve the headline. Qualification: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify **evidence**. Exit: accept a

concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Manuscript says 12 evaluated cases, raw record has 10 completed and 2 missing with 8 passes. Correct the summary.

Teacher explanation: 8/10=**80%** completed pass, coverage10/12 and 2 unassessed; state all denominators. Do not imply12 observed outcomes.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: 8/10=**80%** completed pass, coverage10/12 and 2 unassessed; state all denominators. Do not imply12 observed outcomes. Require intermediate reasoning and the stated scope. Fresh retry: Figure labels old corpus version while methods uses new one. Expected reasoning: Trace which run generated it, correct the version linkage and rebuild mismatched figures if needed. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Manuscript says 12 evaluated cases, raw record has 10 completed and 2 missing with 8 passes. Correct the summary. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Presentation and challenge session. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	

Record	Teacher evidence
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of final project manuscript

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Abstract says 100% success; results table shows 18/20, and two failures are omitted from the figure. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Correct to 90% over 20 with two failures; restore figure and align abstract/table/run. Do not hide failures to preserve the headline.

Check every learner before discussion: Use the worked case for Final project manuscript. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify **evidence**.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The abstract calls the study a test on ten independent plants. Require a **prediction** and a reason before comparing. Expected result: Correct it to five plants with two views each. Reason: Five source plants have two views each; independence requires a justified sampling design.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

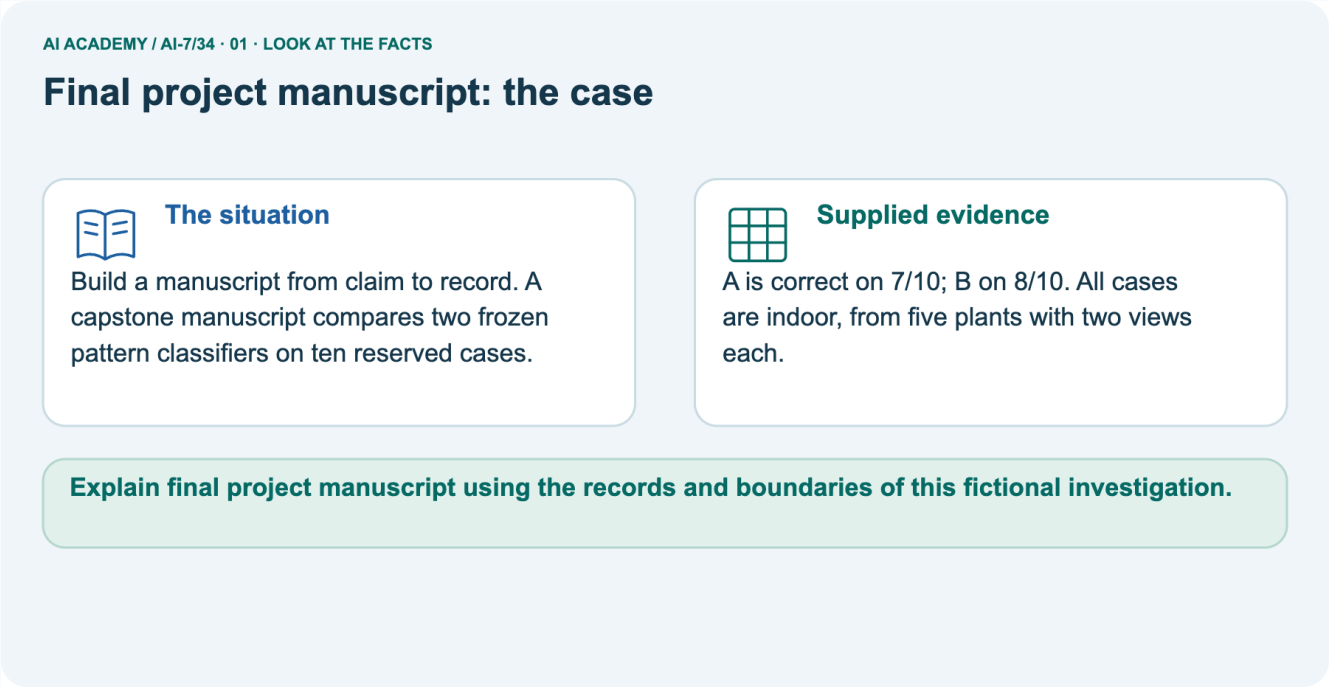
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Final project manuscript: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

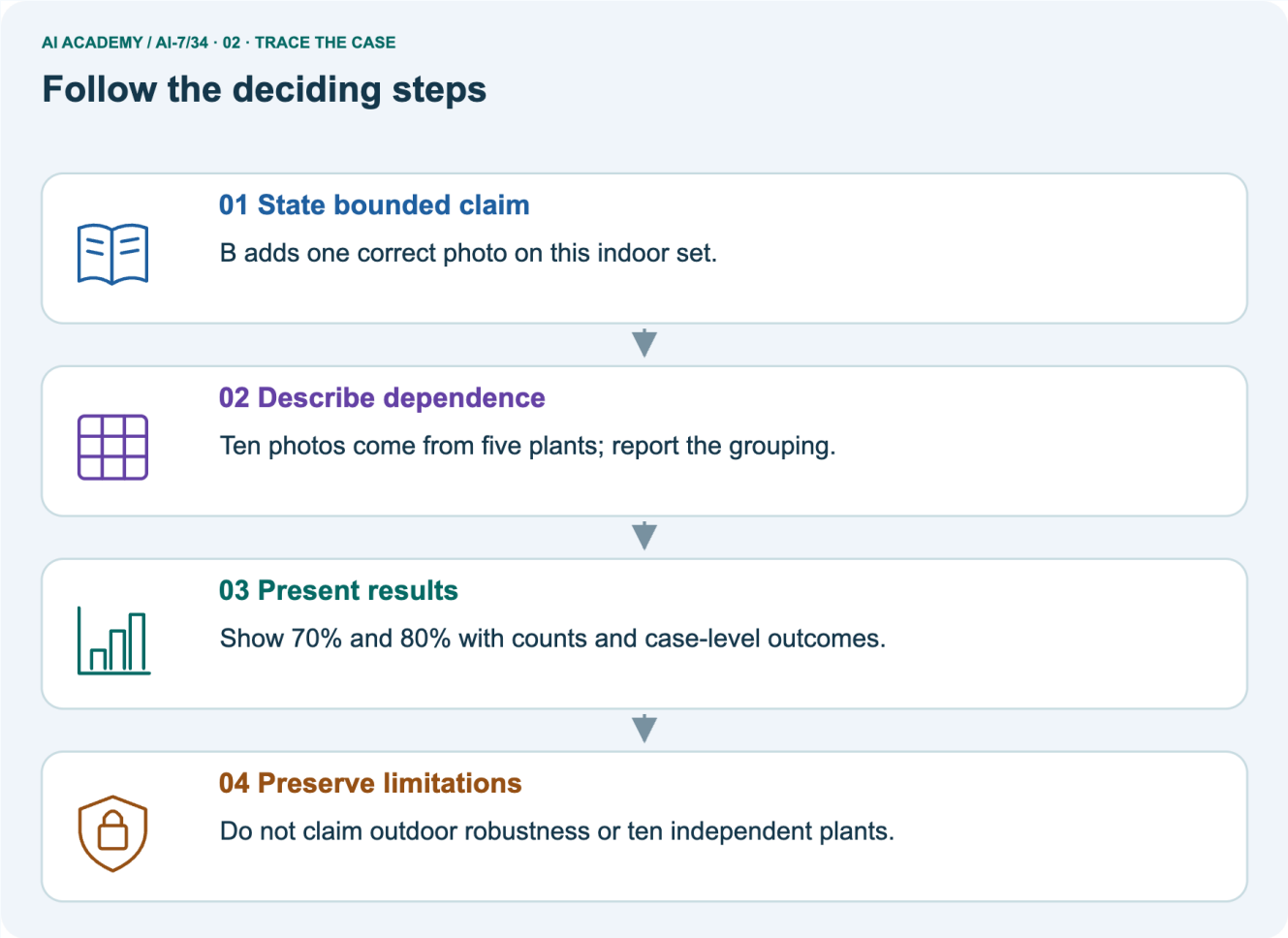
Build a manuscript from claim to record. A capstone manuscript compares two frozen pattern classifiers on ten reserved cases. Supplied facts: A is correct on 7/10; B on 8/10.

All cases are indoor, from five plants with two views each.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

State bounded claim: B adds one correct photo on this indoor set. → Describe dependence: Ten photos come from five plants; report the grouping. → Present results: Show **70%** and **80%** with counts and case-level outcomes. → Preserve limitations: Do not claim outdoor robustness or ten independent plants.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Indoor capstone photo outcomes



Follow the labelled picture with the matching explanation. Open the vector diagram

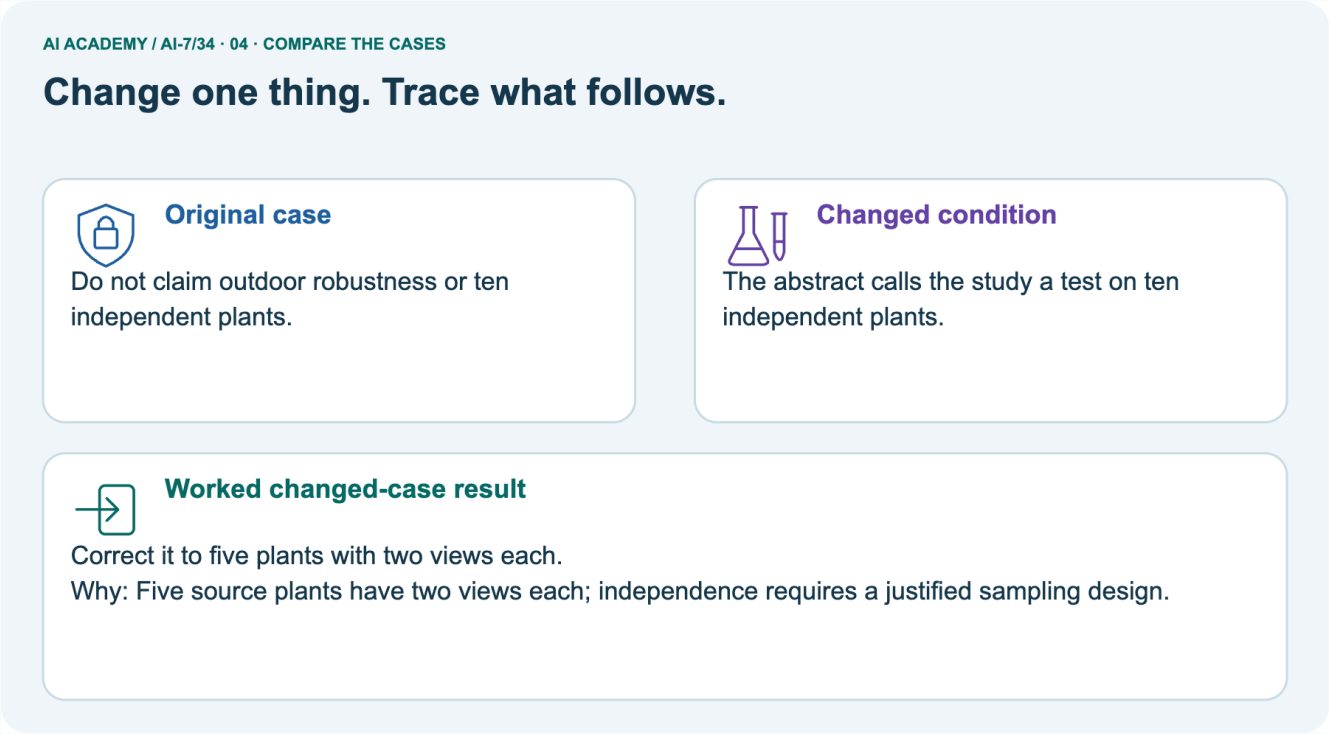
Indoor capstone photo outcomes. Units: correct photos out of 10; five plants. A: 7. B: 8.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



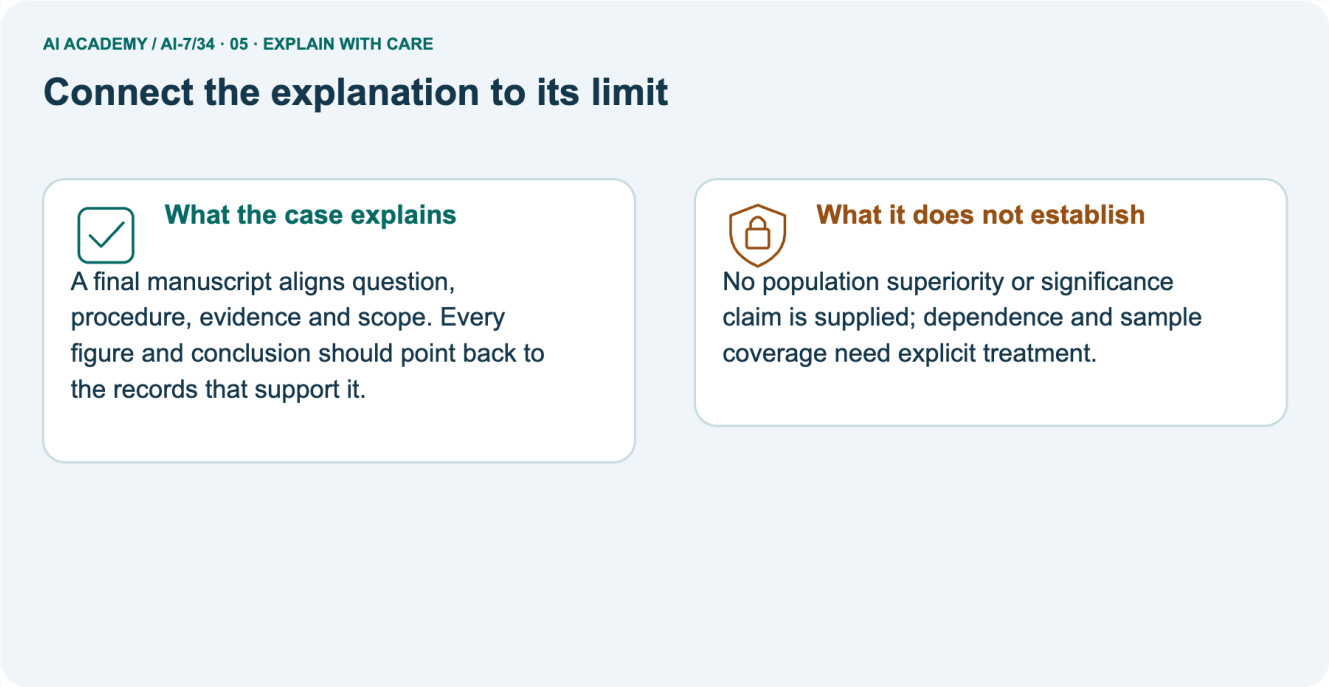
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Do not claim outdoor robustness or ten independent plants. Changed condition: The abstract calls the study a test on ten independent plants. Expected result: Correct it to five plants with two views each. Reason: Five source plants have two views each; independence requires a justified sampling design.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Five source plants have two views each; independence requires a justified sampling design.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A final manuscript aligns question, procedure, **evidence** and scope. Every figure and conclusion should point back to the records that support it.

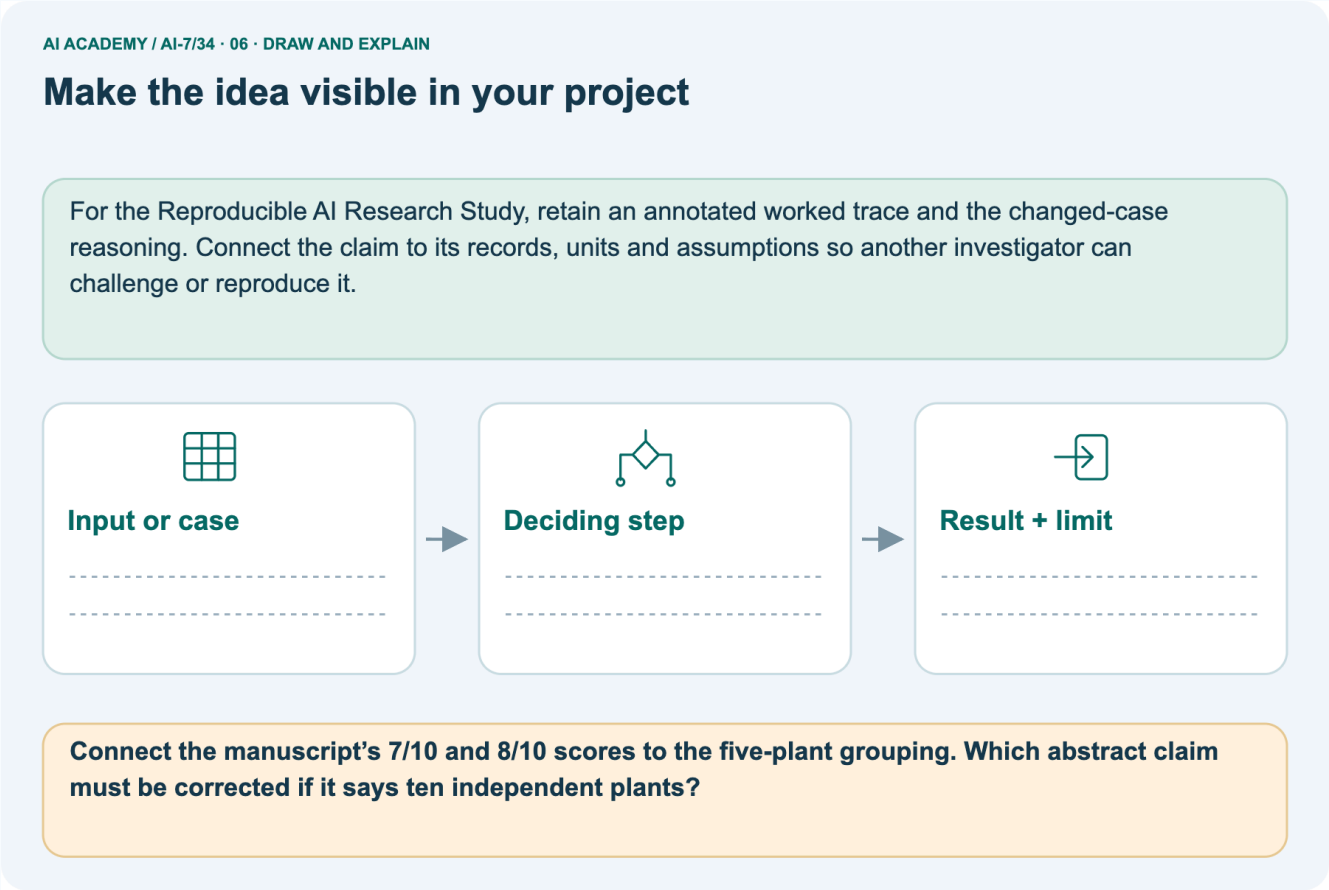
Limit: No population superiority or significance claim is supplied; dependence and sample coverage need explicit treatment.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: No population superiority or significance claim is supplied; dependence and sample coverage need explicit treatment.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Connect the manuscript's 7/10 and 8/10 scores to the five-plant grouping. Which abstract claim must be corrected if it says ten independent plants?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see. Supported practice key: B adds one correct photo on this indoor set. Ten photos come from five plants; report the grouping. Show **70%** and **80%** with counts and

case-level outcomes. Do not claim outdoor robustness or ten independent plants. Changed case: Correct it to five plants with two views each. Five source plants have two views each; independence requires a justified sampling design.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Final project manuscript: the case

02. Follow the deciding steps

03. Indoor capstone photo outcomes

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.